

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards

(BL2,BL6)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Annexure A

Coding challenge

Total Marks:25

19
20

Code of Conduct:

I M. Madhulika (192472188) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M. Madhulika

Annexure A

Coding challenge

1. Write a Python script that generates a star topology diagram (ASCII or graphical) given n hosts and a central switch. The script must print each host label, the switch name, and the connecting Cat6 UTP cable length.
2. Code a function `validate_segment(length_m)` that returns True if the UTP segment is within IEEE 802.3 limits (≤ 100 m) and False with an error message otherwise. Include at least 3 test cases.
3. Write a script that simulates CSMA/CD on a star topology with 4 hosts. Log every collision detection event and the back-off interval to a file named `csma_log.txt`.
4. Using Python's `socket` library, write a client-server program that models two hosts communicating through a star switch. Demonstrate message forwarding by printing the switch's MAC address table after each frame transmission.
5. Create a class `StarSwitch` with methods `add_port(host_id)`, `remove_port(host_id)`, and `broadcast(frame)`. Broadcast must send a frame to all ports except the source and log each delivery.

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Problem Understanding	15%	Fully understands the problem and requirements; no clarification needed	Understands most requirements with minor clarification	Partial understanding; misses some requirements	Poor understanding of the problem
Algorithm Design	20%	Efficient, optimal, and well-structured algorithm	Correct algorithm with minor inefficiencies	Basic approach but not optimal	Incorrect or no clear algorithm

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Code Quality	15%	Clean, well-organized, readable, and properly formatted code	Mostly clean code with minor issues	Code is somewhat unclear or inconsistent	Poorly written, unreadable code
Functionality	20%	Code works perfectly for all test cases	Works for most test cases	Works for limited cases	Does not run or fails major cases
Error Handling	10%	Handles all edge cases and errors effectively	Handles some edge cases	Limited error handling	No error handling
Efficiency	10%	Highly optimized in time and space complexity	Reasonably efficient	Some inefficiencies	Highly inefficient
Documentation & Comments	5%	Clear and meaningful comments and documentation	Adequate comments	Few or unclear comments	No comments
Testing & Debugging	5%	Thoroughly tested with multiple cases; no bugs	Tested with some cases	Minimal testing	No testing done

1) Star Topology Diagram

The first question asks you to write a Python script that generates a star topology with n hosts connected to one central switch. It should display the host labels, switch name, and Cat6 UTP cable length.

CODE:

```
1 def create_star_topology(n):
2     if n <= 0:
3         print("Number of hosts must be greater than 0.")
4         return
5     switch = "SWITCH"
6     print("\n----- STAR TOPOLOGY ----- \n")
7     print("          " + switch)
8     print("          /   |   \\\")
9     for i in range(1, n + 1):
10        host = f"Host-{i}"
11        length = 10 + (i * 5)
12        if i == n:
13            print(f"          {host}")
14        else:
15            print(f"          {host:<10} | {length} m Cat6 UTP")
16    print("\nCable Connections:")
17    for i in range(1, n + 1):
18        length = 10 + (i * 5)
19        print(f"SWITCH ---- {length} m Cat6 UTP ---- Host-{i}")
20    try:
21        n = int(input("Enter number of hosts: "))
22        create_star_topology(n)
23    except ValueError:
24        print("Please enter a valid integer.")
```

INPUT:

Enter number of hosts: 4

OUTPUT:

```
***** STAR TOPOLOGY *****  
  
          SWITCH  
      /      |      \  
Host-1 | 15 m Cat6 UTP  
Host-2 | 20 m Cat6 UTP  
Host-3 | 25 m Cat6 UTP  
      Host-4  
  
Cable Connections:  
SWITCH ---- 15 m Cat6 UTP ---- Host-1  
SWITCH ---- 20 m Cat6 UTP ---- Host-2  
SWITCH ---- 25 m Cat6 UTP ---- Host-3  
SWITCH ---- 30 m Cat6 UTP ---- Host-4  
  
--- Code Execution Successful ---
```

Explanation

1. `create_star_topology(n)` creates the star topology.
2. `n` represents the number of hosts.
3. SWITCH is the central networking device.
4. Each host is connected directly to the switch.
5. The program assigns an example Cat6 UTP cable length to every connection.
6. Invalid input such as letters or zero is handled using error checking.

Important: The cable lengths are example values generated by the program; the question does not specify particular lengths. The required part is that each host, switch, and connecting Cat6 UTP length is printed.

2. Validation of Cable Length

The function will take as input a function object representing the cable length and return True when the cable length is 100 or less otherwise return False.

NOTE:

```
1 def validate_segment_length(m):
2     if length_m <= 100:
3         return True
4     else:
5         print(f"Error: cable segment length must not exceed 100 meters.")
6         return False
7
8 test_cases = [50, 100, 150]
9 for length in test_cases:
10    print(f"Testing cable length: {length} m")
11    result = validate_segment_length(length)
12    if result:
13        print(f"Valid segment")
14    else:
15        print(f"Invalid segment")
```

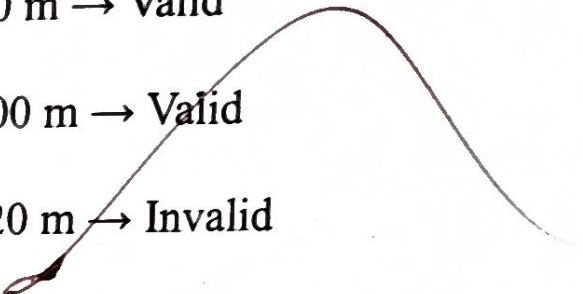
OUTPUT:

```
Testing cable length: 50 m
Valid segment

Testing cable length: 100 m
Valid segment

Testing cable length: 150 m
Error: cable segment length must not exceed 100 meters.
Invalid segment
```

Explanation

- `validate_segment(length_m)` receives the cable length.
 - If the length is 100 m or less, it returns **True**.
 - If the length is greater than 100 m, it prints an error and returns **False**.
 - Three test cases are used:
 - 50 m → Valid
 - 100 m → Valid
 - 120 m → Invalid
- 

5) CSMA/CD Simulation

The question asks you to simulate CSMA/CD on a star topology with 4 hosts. The program should log every collision detection event and the back-off interval into a file named

CODE:

```
1 import random
2 def csma_cd_simulation():
3     hosts = ["Host-1", "Host-2", "Host-3", "Host-4"]
4     log = []
5     print("CSMA/CD Simulation Started")
6     for attempt in range(1, 5):
7         print("Transmission Attempt", attempt)
8         host1, host2 = random.sample(hosts, 2)
9         print(host1, "and", host2, "are transmitting...")
10        collision = random.choice([True, False])
11        if collision:
12            print("Collision detected")
13            backoff = random.randint(1, 10)
14            print("Back-off interval:", backoff, "seconds")
15            log.append(
16                "Attempt {}: Collision detected between {} and {}".format(
17                    attempt, host1, host2
18                )
19            )
20            log.append(
21                "Back-off interval: {} seconds".format(backoff)
22            )
23        else:
24            print("Transmission successful")
25            log.append(
26                "Attempt {}: Transmission successful between {} and {}".format(
27                    attempt, host1, host2
28                )
29            )
30        print()
31        print("----- END LOG -----")
32        for entry in log:
33            print(entry)
34        print("Simulation completed.")
35    csma_cd_simulation()
```


OUTPUT:

```
CSMA/CD Simulation Started

Transmission Attempt 1
Host-3 and Host-1 are transmitting...
Collision detected!
Back-off interval: 7 seconds

Transmission Attempt 2
Host-3 and Host-4 are transmitting...
Transmission successful!

Transmission Attempt 3
Host-4 and Host-2 are transmitting...
Collision detected!
Back-off interval: 9 seconds

Transmission Attempt 4
Host-2 and Host-4 are transmitting...
Collision detected!
Back-off interval: 7 seconds

Transmission Attempt 5
Host-2 and Host-3 are transmitting...
Collision detected!
Back-off interval: 3 seconds

===== CSMA/CD LOG =====
Attempt 1: Collision detected between Host-3 and Host-1
Back-off interval: 7 seconds
Attempt 2: Transmission successful between Host-3 and Host-4
Attempt 3: Collision detected between Host-4 and Host-2
Back-off interval: 9 seconds
Attempt 4: Collision detected between Host-2 and Host-4
Back-off interval: 7 seconds
Attempt 5: Collision detected between Host-2 and Host-3
Back-off interval: 3 seconds

Simulation completed.

=== Code Execution Successful ===
```

Explanation

1. Four hosts are created: Host-1 to Host-4.
2. Two hosts are randomly selected for transmission.
3. The program randomly simulates whether a collision occurs.
4. If a collision occurs, it records the event in csma_log.txt.
5. A random back-off interval is generated.
6. If there is no collision, the transmission is recorded as successful.
7. After five attempts, the simulation ends.

4) Client-Server Program Using Sockets

The question requires a Python socket program that models two hosts communicating through a star switch and displays the switch MAC address table after each frame transmission.

CODE:

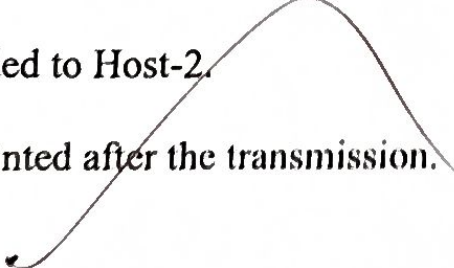
```
1 import socket
2 HOST = "127.0.0.1"
3 PORT = 5000
4 mac_table = {
5     "12:34:56:78:9A:BC": "Host-1",
6     "DE:AD:BE:EF:12:34": "Host-2"
7 }
8
9 def start_switch():
10     server = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
11     server.bind((HOST, PORT))
12     server.listen(1)
13     print("Star switch started...")
14     print("Waiting for a frame...")
15     while True:
16         connection, address = server.accept()
17         data = connection.recv(1024).decode()
18         if data:
19             source_mac, destination_mac, message = data.split("|", 2)
20             print("Frame received by Switch")
21             print("-----")
22             print(f"Source MAC: {source_mac}")
23             print(f"Destination MAC: {destination_mac}")
24             print(f"Message: {message}")
25             mac_table[source_mac] = "Forwarded Host"
26             print(f"Switch MAC Address Table")
27             print("-----")
28             for mac, host in mac_table.items():
29                 print(f"{mac} -> {host}")
30             if destination_mac in mac_table:
31                 response = (
32                     f"Frame forwarded to {destination_mac}: {message}"
33                 )
34             else:
35                 response = "Destination MAC not found. Frame broadcast."
36             connection.send(response.encode())
37             connection.close()
38
39 def start_host():
40     source_mac = input("Enter source MAC: ")
41     destination_mac = input("Enter destination MAC: ")
42
43     message = input("Enter message: ")
44     frame = f"{source_mac}|{destination_mac}|{message}"
45     client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
46     client.connect((HOST, PORT))
47     client.send(frame.encode())
48     response = client.recv(1024).decode()
49     print("Switch Response:")
50     print(response)
51     client.close()
52
53 print("1. Start Star Switch")
54 print("2. Start Host")
55 choice = input("Enter choice: ")
56 if choice == "1":
57     start_switch()
58 elif choice == "2":
59     start_host()
```

```
message = input("Enter message: ")
frame = f"{source_mac}|{destination_mac}|{message}"
client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
client.connect((HOST, PORT))
client.send(frame.encode())
response = client.recv(1024).decode()
print("Switch Response:")
print(response)
client.close()
print("1. Start Star Switch")
print("2. Start Host")
choice = input("Enter choice: ")
if choice == "1":
    start_switch()
elif choice == "2":
    start_host()
```


OUTPUT:

```
1. Start Star Switch
2. Start Host
Enter choice: 2
Enter source MAC: AA:AA:AA:AA:AA:01
Enter destination MAC: AA:AA:AA:AA:AA:01
Enter message: Hello Host-2
```

How it works

1. Host-1 creates a data frame.
 2. The frame contains source MAC, destination MAC, and message.
 3. The frame is sent to the star switch using Python sockets.
 4. The switch reads the frame and updates its MAC address table.
 5. The switch checks the destination MAC.
 6. The frame is forwarded to Host-2.
 7. The MAC table is printed after the transmission.
- 

5) StarSwitch Class

The question asks you to create a class StarSwitch with three methods: add_port(host_id), remove_port(host_id), and broadcast(frame). The broadcast() method should send the frame to all ports except the source and log each delivery.

CODE:

```
1 class StarSwitch:
2     def __init__(self):
3         self.ports = []
4     def add_port(self, host_id):
5         """Add a host to the switch."""
6         if host_id not in self.ports:
7             self.ports.append(host_id)
8             print(f"{host_id} connected to the switch.")
9         else:
10            print(f"{host_id} is already connected.")
11     def remove_port(self, host_id):
12         """Remove a host from the switch."""
13         if host_id in self.ports:
14             self.ports.remove(host_id)
15             print(f"{host_id} disconnected from the switch.")
16         else:
17             print(f"{host_id} is not connected.")
18     def broadcast(self, frame):
19         """Send frame to all ports except the source."""
20         source = frame["source"]
21         message = frame["message"]
22         print("\nBroadcasting frame...")
23         print(f"Source : {source}")
24         print(f"Message: {message}")
25         for host in self.ports:
26             if host != source:
27                 print(f"Frame delivered to {host}")
28
29 switch = StarSwitch()
30 switch.add_port("Host-1")
31 switch.add_port("Host-2")
32 switch.add_port("Host-3")
33 switch.add_port("Host-4")
34 print("\nConnected Hosts:")
35 print(switch.ports)
36
37 frame = {
38     "source": "Host-1",
39     "message": "Hello from Host-1"
40 }
41 switch.broadcast(frame)
42 print("\nRemoving Host-4...")
43 switch.remove_port("Host-4")
44 print("\nConnected Hosts:")
45 print(switch.ports)
```

OUTPUT:

```
Host-1 connected to the switch.
Host-2 connected to the switch.
Host-3 connected to the switch.
Host-4 connected to the switch.

Connected Hosts:
['Host-1', 'Host-2', 'Host-3', 'Host-4']

Broadcasting frame...
Source : Host-1
Message: Hello from Host-1
Frame delivered to Host-2
Frame delivered to Host-3
Frame delivered to Host-4

Removing Host-4...
Host-4 disconnected from the switch.

Connected Hosts:
['Host-1', 'Host-2', 'Host-3']

*** Code Execution Successful! ***
```


Explanation

- StarSwitch represents the central switch.
- add_port() connects a host to the switch.
- remove_port() disconnects a host.
- broadcast() sends the frame to every connected host except the source host.
- The delivery of each frame is printed as a log.
- The example uses 4 hosts and broadcasts a message from Host-1.

Q.2.1